

Response to the editor and reviewers

Alexander V. Mantzaris

This response concerns submission 0170c7b2-1065-4344-a5d5-b839b51ac22c, *RankCloak Conceals the Surface Form of Synthetic Cryptographic Artifacts in Language Model Generated Text*. The original review wording is quoted in full. A new intact-message study supplies automated contextual evidence for Reviewer 1 comment 4. Human perception remains unmeasured. Page references identify the accompanying manuscript and Supplementary Information, with stable section, figure and table names for other renderings.

Editor letter

— Dear Dr Mantzaris,

Your manuscript, "*RankCloak Conceals the Surface Form of Synthetic Cryptographic Artifacts in Language Model Generated Text*", has now been assessed.

We invite you to revise your paper, carefully addressing the comments from the reviewers and the editor. Please ensure the results are accurately reported, any overstated conclusions are rewritten and the limitations of the work fully explained. When your revision is ready, please submit the updated manuscript and a point-by-point response. This will help us move to a swift decision.

Please note that if your manuscript uses any custom or bespoke computational tool or code, or reports a new algorithm, tool, software, or a pipeline (even if individual components are not new), the underlying code must be deposited in a recognised DOI-assigning repository (e.g. zenodo) and linked either from Methods or a dedicated Code Availability section.

Editor Comments

"The manuscript presents a technically interesting and systematically evaluated framework for embedding cryptographic payloads into language-model-generated text through deterministic token-rank manipulation. Reviewer 2 considers the previous concerns largely addressed. However, several substantive issues remain before the conclusions can be fully supported. In particular, the authors should clarify the robustness boundaries under text transformations and configuration desynchronization, analyze the implications of zero cross-model recovery, explain the near-perfect detectability under an exact-model-aware attacker, strengthen semantic-coherence validation, document all deterministic filtering rules, and provide deeper analyses of entropy-gating failures, computational complexity, and quantization-induced divergence. Suggested references are optional, and the authors may select appropriate recent peer-reviewed literature." -Chun-Wei Yang

We recommend submitting all revisions within the mentioned deadline.

If you need more time, please contact us and include your submission ID.

Kind regards,

Purva Dhamankar —

Accuracy and scope.

Answer. We have revised the claims to distinguish surface-form concealment, exact configured recovery, visible-text transport, detectability and reader-perceived naturalness. Adverse and inconclusive outcomes remain visible.

Change and evidence. The V4 text corrects the historical raw and limited unmodified references to saved-token recovery, documents the unfiltered sampled-skip entropy experiment, and separates the directional fixed-message quantization endpoint from free generation. The new contextual follow-up uses matched greedy controls and reports automated rubric acceptance separately from human perception. All earlier fragment outcomes remain in the supplement.

Location. Abstract, main Results under Contextual acceptability of complete messages, Discussion, and Supplementary Notes S16 through S20. Accompanying documents, main4.pdf pages 11, 16, 19 and supplementary4.pdf pages 41, 42.

Code deposit.

Answer. The code and retained data for this V4 revision are available at <https://github.com/mantzaris/llm-rankcloak>. Published version 2.0.0, DOI 10.5281/zenodo.22555497, covers V3 only.

Change and evidence. The availability statements distinguish repository access to V4 from the published V3 archive. A new version under concept DOI 10.5281/zenodo.21987449 covering the current V4 code and evidence remains outstanding.

Location. Main Data Availability and Code Availability. Accompanying documents, main4.pdf pages 20.

Technical assessment.

Answer. The revision retains the theoretical and implementation analyses and limits the contribution to artifact-specific surface-form concealment and configured inversion. A new intact-message study now supplies the primary automated contextual evidence. Automated contextual evidence remains distinct from unmeasured human perception.

Change and evidence. The new study applies the same lenient, fixed rubric to complete encoded messages and matched greedy ordinary controls. It uses authentic prompts, blinded cross-family judges, a separate constructed calibration set and a frozen 48-payload sample. Among 537 complete message pairs in 280 trials, the equally weighted panel accepts 90.3 [88.4, 92.1]% of encoded messages and 96.6 [95.0, 98.1]% of matched ordinary messages. The paired difference is -6.3 [-8.7, -3.9] percentage points. Brackets give 95% class-stratified payload-cluster intervals. Mean logical connectedness is 3.60/5 for encoded text and 4.00/5 for ordinary text. Binary judge disagreement is 14.6% and 6.7%, respectively. The previous fragment assays are retained as supplementary methodological history, with their failed semantic control checks and limited local likelihood findings. Historical recovery, transport, entropy, quantization and detector results are unchanged. No reader validation, confidentiality or indistinguishability is claimed.

Location. Abstract, contextual Methods and Results, Discussion and limitations, Supplementary Notes S17 and S20, and the answer to Reviewer 1 comment 4. Accompanying documents, main4.pdf

Reviewer 2 acknowledgment

Reviewer 2

Thank the authors for addressing most of my concerns.

Answer. We thank Reviewer 2 for acknowledging the earlier revisions.

Change and evidence. The acknowledgment is retained in full with the current reviewer label. This revision responds separately to the new editor and Reviewer 1 requests.

Location. This response, Reviewer 2 acknowledgment. No manuscript change is needed for this acknowledgment.

Reviewer 1 general assessment and wording instruction

Reviewer 1

The manuscript proposes a steganographic method embedding cryptographic payloads into language model outputs via deterministic token rank manipulation. While demonstrating exact recovery under strictly controlled conditions, the framework exhibits fragility against text transformations and cross-model decoding, limiting its practical utility despite comprehensive evaluations of capacity and neural steganalysis detection.

Overall, the study presents a reproducible measurement framework for linguistic steganography. However, its reliance on perfectly synchronized configurations and failure during standard markdown operations require deeper theoretical justification. Furthermore, near-perfect detectability by exact-model-aware attackers and the absence of perceptual validation weaken its broader claims. Significant revisions addressing robustness boundaries, theoretical limitations of detectability, and expanded discussions on quantization sensitivity are required before publication can be considered.

Please do not arbitrarily simplify or rephrase the review comments, as this hinders the assessment of your revisions.

General assessment.

Answer. We agree that recovery requires a synchronized configuration and that the observed transport and detection failures substantially restrict practical claims. The revision presents these as limitations, not as problems solved by a new robustness method.

Change and evidence. The discussion states that a fully configured attacker may decode, distinguishes trace-informed detection from visible-text scoring, and labels the block/ECC fallback as unimplemented. A new intact-message contextual evaluation supplies automated evidence under an explicit lenient rubric. The unsuccessful earlier semantic pilots remain in Supplementary Note S17. The additional transport diagnostics are illustrative rather than population estimates.

Location. Abstract, main Discussion under Configuration sensitivity, detection and a proposed fallback, limitations, and Supplementary Notes S17 through S20. Accompanying documents, main4.pdf pages 19, 20 and supplementary4.pdf pages 36, 40, 41, 42.

Preservation of the review wording.

Answer. We have retained the complete review wording without simplification or paraphrase, including both general paragraphs and this instruction.

Change and evidence. All thirteen original quoted blocks are retained, including the editor letter, Reviewer 2 acknowledgment, both Reviewer 1 introductory paragraphs and the instruction to preserve wording. Answers remain separate from the original comments.

Location. All review-comment blocks in this response. The original requests file and exact-block ledger remain unchanged.

Reviewer 1 comment 1

1- In the abstract and introduction, the manuscript admits severe transmission fragility. While you explicitly scope this as a measurement framework, you must expand the theoretical discussion on why standard operations like markdown copy cause complete failure and discuss potential boundary mitigations.

Answer. The tested Markdown operation is a specific composite transform. It adds a blockquote prefix to each line and removes whitespace. The receiver then applies saved token offsets after retokenization. This can shift the forced span, change tokens and alter contextual ranks. It does not demonstrate failure for every Markdown workflow.

Change and evidence. Five hash-selected sources give 25 decomposed arms and 135 segment instances. Exact reuse leaves 26 missing segment replays. We also checked all 53 reused segment inputs at the new context allocation, with identical rank and error outcomes. These checks add no logical cover arms. Prefix-only insertion recovers none of the five. All 27 selected segment instances are rejected at a prohibited first forced token before a rank sequence is returned. These endpoints do not measure downstream rank amplification. Declared wrapper removal recovers one originally failing Mistral case, but cannot restore removed whitespace. Byte edits, extracted spans, first rank differences, errors and recovered bytes are retained. Original composite recovery remains 0/144.

Location. Main Methods under Decoder compatibility and bounded transport diagnostics, Results under Mechanisms of transmission failure, and Supplementary Note S18 with its illustrative recovery table. Accompanying documents, main4.pdf pages 10, 18 and supplementary4.pdf pages 40, 41.

Reviewer 1 comment 2

2- Within the bounded payload methods section, assuming perfectly synchronized model configurations is highly restrictive. Please introduce a theoretical discussion addressing how minor configuration desynchronization, such as slight tokenizer version updates, theoretically impacts the overall payload recovery rate.

Answer. A minor version change has no universal recovery-rate implication. It may preserve the relevant tokenization or change token identities, offsets and conditioning. Model or gate differences introduce other disagreement mechanisms. We state the required shared contract and conditional agreement argument explicitly.

Change and evidence. A new canonical configuration fingerprint includes eleven required contract components. Tests accept matching configurations, verify deterministic serialization and reject

changes to every component. It is a compatibility API, not authentication or a robustness guarantee, and was not retroactively applied to the historical messages.

Location. Main Methods under Decoder compatibility and bounded transport diagnostics, main Discussion under Configuration sensitivity, detection and a proposed fallback, and Supplementary Note S19. Accompanying documents, main4.pdf pages 10, 19 and supplementary4.pdf pages 41.

Reviewer 1 comment 3

3- In the exact copy fragility evaluation, the zero percent recovery rate for cross-model decoding is a severe limitation. You should propose and discuss a conceptual error correction strategy or fallback mechanism to mitigate this lack of robustness during mismatched decoding.

Answer. Zero cross-model recovery remains a severe limitation. We propose configuration identification before decoding, independently framed blocks with context resets, integrity checks, bounded outer error/erasure correction and retransmission. This is a conceptual fallback, not an implemented improvement.

Change and evidence. The argument states the minimum-distance condition $2e+s<d$ and its assumptions. ECC cannot reconstruct an arbitrary wrong model codebook or repair persistent synchronization loss. Reliable framing, a supported configuration and overhead for redundancy and retries remain necessary. Retransmission under an unresolved mismatch is ineffective.

Location. Main Discussion under Configuration sensitivity, detection and a proposed fallback, and Supplementary Note S19. Accompanying documents, main4.pdf pages 19 and supplementary4.pdf pages 41.

Reviewer 1 comment 4

4- Throughout the segmented multi-cover method section, the distinction between payload-bearing spans and natural tails lacks perceptual validation. You must provide stronger automated semantic coherence metrics to prove that these generated transitions appear natural and logically connected to readers.

Answer. We have made intact-message contextual acceptability the primary automated coherence evaluation. Two blinded local model judges assess each complete message in its authentic prompt context without seeing the payload boundary. The fixed rubric accepts clear meaning and minor awkwardness at disruption levels 0 or 1, while levels 2 and 3 indicate material interference with understanding. The study was designed after the fragment results and frozen before its own scoring.

Change and evidence. A seeded selection covers 48 payloads across four hexadecimal classes, three generators and two schedules, giving 576 messages outside the earlier scored cohort. Each is paired with ordinary output from the same generator and prompt using matched filtering, greedy continuation, budget and stopping rules. Among 537 complete pairs, panel acceptance is 90.3% for RankCloak and 96.6% for ordinary output. The paired difference is -6.3 percentage points with a 95% payload-cluster interval of [-8.7, -3.9]. Mean logical connectedness is 3.60/5 versus 4.00/5.

Messages are averaged within trials and payloads, with intervals conditional on the fixed reused controls. Supplementary Note S20 reports calibration mistakes, the 39 incomplete pairs, missing-

score sensitivity, subgroup results, disagreement and exact unfavorable examples. Judges and generators are not fully crossed, and the comparison does not isolate a causal boundary effect. This provides automated evidence of contextual acceptability, not measured human perception or indistinguishability. The earlier fragment pilots and limited likelihood evidence remain in S17. The editor can assess whether this evidence meets the requested validation.

Location. Main Methods, Intact-message contextual acceptability, and Results, Contextual acceptability of complete messages, with Table 3 and Figure 5. Supplementary Note S20, Selection and matched generation, Complete judging rubric, Judge settings and calibration, Estimation and complete results, Stops, lengths, reuse and missingness, and Full-message examples. Note S17 retains the historical assays. Accompanying documents, main4.pdf pages 10, 16, 17 and supplementary4.pdf pages 36, 42, 44, 45, 46.

Reviewer 1 comment 5

5- Neural steganalysis results indicate near-perfect detection by the exact-model-aware attacker. Rather than a full redesign, you must deeply analyze the theoretical bottlenecks causing this glaring statistical detectability and clearly define the framework’s security boundaries against this specific threat model.

Answer. The evaluated model-aware attack is trace informed and exposes a strong statistical limitation. Its features come from saved token likelihoods under the generating model. Reproducing them requires the original token path, model/backend, prompts, boundaries, resets and probability convention. A visible-text-only scorer is a distinct condition.

Change and evidence. The added discussion explains that payload-rank distributions need not match ordinary sampling. It distinguishes untempered full-vocabulary scores from the temperature-0.8 top-p-0.95 sampler and states the support conditions for a sequence-KL argument. AUC is not converted to KL, and no unique causal feature attribution is claimed. An attacker with the complete receiver configuration may decode. No confidentiality or informed-attacker indistinguishability is asserted.

Location. Main Neural steganalysis, statistical analysis, and computational overhead, Configuration sensitivity, detection and a proposed fallback, and Supplementary Notes S12 and S19. Accompanying documents, main4.pdf pages 9, 19 and supplementary4.pdf pages 23, 41.

Reviewer 1 comment 6

6- The discussion of entropy gating notes that the strict gate condition resulted in maximum-budget payload completion failures. You should investigate and report the exact linguistic contexts or token-level characteristics that triggered these specific failures during the generation loop.

Answer. All six strict-gate failures are retained and analyzed at their original budgets. Five are Qwen and one Mistral, all ASCII B16. They consume 110/128, 116/128, 54/64, 106/128, 59/128 and 116/128 ranks. Strict completion remains 114/120.

Change and evidence. The supplement reports exact pinned-tokenizer windows, longest below-threshold runs of 50, 163, 39, 35, 267 and 51 positions, rolling progress, margins and rank pressure. The longest runs contain 30 to 157 distinct token IDs, at most three identical IDs consecutively,

and rank-one frequencies of 85.7 to 98.8 percent. All 605 positions remain ordinary sampled skips with no payload progress. Twelve matched ungated/moderate trajectories and three hash-selected strict successes are retained. Observed contexts include formatted public-service and project-update passages, but low-entropy runs also occur in successes. These are descriptive associations, not established linguistic causes. The trajectories used no token filter and sampled ordinary skips, not greedy skips.

Location. Main Results under Strict-gate failure contexts and Supplementary Notes S13 and S16, including the six-case table, trajectory figure and exact local windows. Accompanying documents, main4.pdf pages 18 and supplementary4.pdf pages 26, 33, 35.

Reviewer 1 comment 7

7- The computational overhead analysis reports significant performance bottlenecks. Instead of exhaustive hardware baselines, please provide a theoretical computational complexity analysis comparing your rank transcoding method against standard linguistic steganography tools to mathematically justify the observed processing costs.

Answer. The new comparison separates model work from sorting, rank counting, probability passes and coding operations. Rank selection sorts all allowed candidates, whereas recovery of a known token counts greater logits and lower-ID ties.

Change and evidence. The analysis inspects the primary arithmetic-coding and SAAC implementations and ADG grouping algorithm. It includes repeated diagnostic sorts, finite-precision cumulative tables, grouping search/removal, prompt prefills, KV state, logits_all storage, tails, sampled skips, segmentation and payload-side direct-subword passes. Bounds are implementation-specific and imply no cross-backend wall-time superiority.

Location. Main Computational overhead and generalizability and Supplementary Note S19, implementation-specific operation-count table. Accompanying documents, main4.pdf pages 18 and supplementary4.pdf pages 41, 42.

Reviewer 1 comment 8

8- In the reference list, citation 16 is an unverified preprint. If formally published, update the citation; otherwise, provide detailed theoretical justification for relying on its entropy-guided watermarking claims, or replace it with a peer-reviewed publication to strengthen your foundation.

Answer. The bibliography key Cai2025EntropyGuidedWatermarking refers to the previous reference 16, arXiv:2504.12108. The inspected primary record lists no journal reference. We have replaced its role in the V4 argument with the peer-reviewed Lee and colleagues ACL 2024 paper on entropy-threshold selective watermarking.

Change and evidence. The V4 prose now cites Lee2024SWEET for the limited thresholding precedent. It distinguishes detecting a watermark from recovering a chosen payload. Neither a watermark theorem nor the unverified preprint is used as a RankCloak recovery guarantee. The historical experiment itself is still documented by its original configuration and data.

Location. Main Introduction, Supplementary Note S13 and the V4 bibliography. Accompanying documents, main4.pdf pages 1 and supplementary4.pdf pages 26.

Reviewer 1 comment 9

9- The methods section describing the deterministic token filter leaves the rules for excluding markup fragments obscure. You must explicitly list these complete exclusion criteria in the main text so independent researchers can accurately replicate your experimental setup.

Answer. The main Methods now give the complete deterministic token filter specification, including every literal blocked substring and all other exclusion criteria. The historical filter is unchanged.

Change and evidence. The specification covers empty and replacement pieces, ASCII controls and exceptions, whitespace/start rules, backslashes, individual-token decoding and exceptions, mask application, special-token distinctions and stable tie-breaking. It distinguishes full selection sorting from counted rank recovery and notes limitations for concatenated markup and nonfinite logits. The versioned source link and source hash are retained in the audit, with focused edge-case tests.

Location. Main Methods under Complete deterministic token filter specification and its complete rule table. Accompanying documents, main4.pdf pages 5.

Reviewer 1 comment 10

10- The paired quantization experiment shows independently generated token paths diverging significantly despite identical inputs. Please expand this section to theoretically discuss how minor quantization errors in logits are amplified by the deterministic sorting process, triggering this avalanche effect.

Answer. We separate same-history logit perturbations from subsequent feedback when independently generated token paths split. For a fixed history and allowed set, pairwise gaps above twice a uniform absolute perturbation bound preserve order. Smaller gaps and ties have no such protection. This is a conditional bound, not a measured quantization-error distribution.

Change and evidence. The audit reconciles 69,528 rank changes across 244,440 positions in 1,920 pairs. Encoded messages contribute 56,321/122,220 and ordinary controls 13,207/122,220. The distinct offline fixed-message endpoint gives zero exact Q4-to-Q8 payload recoveries in 960 supported rows, with 766 invalid-rank trials and 194 valid but wrong decoded payloads. It supports no reverse-direction claim. Observed-token replay feeds received cover tokens, not decoded payload symbols, back into the model. Existing evidence supports these claims without new neighboring-logit replay.

Location. Main Results under Fixed-message cross-quantization decoding and Supplementary Notes S14, S18 under Directional cross-quantization endpoint, and S19. Accompanying documents, main4.pdf pages 18 and supplementary4.pdf pages 29, 41.